

CSE 112 – Triangle Class – credit/no credit option

Create a Triangle class in a separate header file that has the following private member variables only:

double side1, side2, side3;

The class will have the following member functions:

1. A Constructor that accepts two (2) sides of the triangle that are not the hypotenuse, and populates side3 (the hypotenuse) by calling the function compute_side3.
2. compute_side3 – returns a double that is the computed length of the 3rd side (the hypotenuse) using the other two sides.
3. compute_perimeter – returns a double that is the perimeter of the triangle.
4. compute_area – returns a double that is the area of the triangle (see Heron's Formula below)

Program Requirements:

The main program provided below will demonstrate the triangle class. The output precision will be set to (2) two decimal places. A UML Diagram for the class will be included as a comment below main.

Heron's formula for the area of a triangle

$$A = \sqrt{s((s-a)(s-b)(s-c))}$$

- In the equation, "s" is half of the perimeter of the triangle.
- For a test using 3 and 6 as the side lengths, the area will be 9.
- For a test using 4.5 and 5.6 as the side lengths, the area will be 12.60.

Comment below main - UML diagram with Access Specifiers (ref pg 801).

```
// *****  
//  CLASS NAME  
// *****  
//  MEMBER VARIABLES with access specifiers  
//  
// *****  
//  MEMBER FUNCTIONS with access specifiers  
//  parameters, and return values  
// *****
```

Grading (credit/no credit) will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation.

Main:

```
#include <iostream>
#include <iomanip>
#include "Triangle.h"

using namespace std;

int main()
{
    double side1, side2, side3;

    cout << "Computes the perimeter of a Triangle from 2 sides." << endl;
    cout << "Enter the length of side 1: ";
    cin >> side1;
    cout << "Enter the length of side 2: ";
    cin >> side2;

    Triangle triangle(side1, side2);

    cout << setprecision(2) << fixed;

    cout << "\nThe side lengths are:" << endl;
    cout << "\t side 1: " << side1 << "\t side 2: " << side2 << "\t side 3: " << triangle.compute_side3();
    cout << endl << endl;
    cout << "The perimeter is: " << triangle.compute_perimeter() << endl << endl;
    cout << "The area of the triangle is: " << triangle.compute_area() << endl;

    return 0;
}
```